TABLEAU PORTFOLIO PROJECT

GROUP C: STORE DISTANCE -- GEOSPATIAL RANGE ANALYSIS

Store Distance & Geospatial Range Analysis

Great-circle distance calculation on a satellite aerial map of New York City

Tableau | Geospatial | Great-Circle Distance | Satellite Map | Custom Shape | Buffer Analysis

Project Summary

A geospatial Tableau view plotting coffee shop locations across Manhattan and surrounding NYC boroughs on a Mapbox satellite tile base map. Each store icon encodes two dimensions: color shows whether the store is within or outside a defined service range (blue = In Range, orange = Out of Range), while size encodes the minimum NY Great-Circle Distance from a reference point (0.0724 to 1.0292 units). Demonstrates custom shape assignment, spatial distance calculation, and satellite map background integration.

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1 PROJECT OVERVIEW

Workbook:	Store Distance -- NYC Geospatial Range Analysis
Views:	1 (Satellite aerial map with proportional coffee shop icons)
Data Source:	NYC store location dataset with lat/long coordinates and service range flags
Map Type:	Mapbox satellite/aerial tile via Tableau Background Maps
Distance Metric:	NY Great-Circle Distance (spherical Haversine calculation)
Mark Type:	Custom shape -- coffee cup icon
Color Encoding:	Out of Range: Blue (No) = In Range, Orange (Yes) = Out of Range
Size Encoding:	MIN(NY Great-Circle Distance) -- 0.0724 to 1.0292
Business Use Case:	Retail site coverage analysis -- identify stores outside service radius

2 BUSINESS CONTEXT

Retail chains, delivery services, and logistics providers need to understand which store locations fall outside a defined service radius -- for staffing, delivery routing, or coverage gap analysis. This workbook builds that analysis in Tableau using great-circle distance calculations on a satellite map base, giving stakeholders an intuitive aerial view of their coverage footprint across the New York City urban landscape.

- Blue coffee cup icons (In Range): Stores within the service radius. The majority of Manhattan and surrounding borough stores fall in this category.
- Orange coffee cup icons (Out of Range): Stores beyond the threshold distance. Clustered primarily along the western Manhattan waterfront and lower Manhattan, suggesting service gaps in high-density zones.
- Icon size gradient: Smaller icons = closer to reference point. Larger icons = further away -- reinforcing spatial distance visually beyond just color.
- Satellite background: Provides real urban context (streets, buildings, waterways) that abstract tile maps cannot -- critical for route planning and physical accessibility decisions.

3 DASHBOARD VIEW -- STORE DISTANCE MAP



Store Distance -- NYC satellite aerial map. Coffee cup icons: blue = In Range, orange = Out of Range. Icon size = MIN(NY Great-Circle Distance), range 0.0724 to 1.0292.

4 TECHNICAL EXPLANATION

The Great-Circle Distance calculation uses the Haversine formula adapted for Tableau calculated fields:

- Haversine formula: d = 2r * arcsin(sqrt(sin^2((lat2-lat1)/2) + cos(lat1)*cos(lat2)*sin^2((lon2-lon1)/2)))
- In Tableau: Built as a calculated field using RADIANS(), SIN(), COS(), SQRT(), ASIN() functions against the store lat/long fields and a fixed reference point parameter.
- The "NY Great-Circle Distance" field normalizes the result to a 0-1+ scale representing distance in degrees or a custom unit.
- Custom shapes: Coffee cup icons assigned via Tableau's Shapes marks card -- requires placing custom PNG icons in the Tableau Repository Shapes folder.
- Out of Range flag: A calculated boolean field [Out of Range] = IF [NY Great-Circle Distance] > [Range Threshold Parameter] THEN 'Yes' ELSE 'No' END -- drives the color encoding.
- Satellite tiles: Selected via Map > Background Maps > Satellite in Tableau Desktop.
- MIN() aggregation on distance: Handles the case where a store appears in multiple rows -- MIN selects the closest measurement per store location.

5 SKILLS DEMONSTRATED

- Great-circle distance calculation: Haversine formula implemented as Tableau calculated field.
- Custom shape assignment: Coffee cup icons via Tableau Repository custom shapes folder.
- Satellite/aerial Mapbox tile integration via Tableau Background Maps.
- Dual-encoding marks: Color (range category) and Size (continuous distance) on same mark.
- Parameter-driven buffer radius: Range threshold adjustable at runtime.
- Spatial coverage gap analysis: Visual identification of service holes in dense urban areas.